

Supplemental theory: a strict Rayleigh bound state in the continuum in a two-groove underwater acoustic metagrating

Author Name¹ and Coauthor Name¹

¹*Affiliation to be inserted*

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SCOPE AND NOTATION

This supplement gives the modal derivation and numerical diagnostics behind the theory-only main text. It does not add an experiment, a finite-element calculation, or a loss model. The water sound speed is $c = 1500$ m/s, the nominal frequency is $f_0 = 200$ kHz, and the time convention is $e^{+i\omega t}$. Superscripts dr and hom distinguish driven scattering amplitudes from outgoing amplitudes of a homogeneous eigenmode. The strict Rayleigh criterion is imposed on the latter.

FLOQUET EXPANSION AND GROOVE MATCHING

For a period a and incidence angle θ , define

$$k_0 = \frac{\omega}{c}, \quad k_{x,\text{inc}} = k_0 \sin \theta, \quad k_{y,\text{inc}} = k_0 \cos \theta. \quad (\text{S1})$$

The incident field travels toward $-y$ and the outgoing reflected expansion above the surface is

$$p_{\text{inc}} = e^{i(k_{x,\text{inc}}x + k_{y,\text{inc}}y)}, \quad p_{\text{sc}} = \sum_n A_n^{\text{dr}} e^{ik_{x,n}x - ik_{y,n}y}, \quad (\text{S2})$$

$$\begin{aligned} k_{x,n} &= k_{x,\text{inc}} + \frac{2\pi n}{a}, \\ k_{y,n} &= \sqrt{k_0^2 - k_{x,n}^2}, \\ \Re k_{y,n} &\geq 0, \quad \Im k_{y,n} \leq 0. \end{aligned} \quad (\text{S3})$$

With $\Omega = a/\lambda$ and $\kappa = ak_{x,\text{inc}}/(2\pi)$, the Rayleigh condition for order n is

$$|\kappa + n| = \Omega. \quad (\text{S4})$$

For groove ℓ , let x_ℓ be the left edge and let the groove occupy $x_\ell < x < x_\ell + w_\ell$ and $-d_\ell < y < 0$. A bottom-rigid cosine expansion is

$$\begin{aligned} p_\ell(x, y) &= \sum_{q=0}^{K-1} C_{\ell q} \cos\left[\frac{q\pi(x - x_\ell)}{w_\ell}\right] \\ &\quad \times \cos[\beta_{\ell q}(y + d_\ell)], \\ \beta_{\ell q}^2 &= k_0^2 - \left(\frac{q\pi}{w_\ell}\right)^2. \end{aligned} \quad (\text{S5})$$

The aperture projection used by the matching calculation is

$$\Pi_{n,\ell q} = \frac{1}{w_\ell} \int_{x_\ell}^{x_\ell + w_\ell} e^{-ik_{x,n}x} \cos\left[\frac{q\pi(x - x_\ell)}{w_\ell}\right] dx. \quad (\text{S6})$$

The cosine norm is 1 for $q = 0$ and $1/2$ for $q > 0$; the corresponding factors are included in the numerical projection matrices.

For the chosen time convention, pressure-normalized normal velocity is $\rho\omega v_y = +i\partial_y p$. At the groove aperture this gives

$$\rho\omega v_{y,\ell}(x, 0) = -i \sum_q \beta_{\ell q} C_{\ell q} \sin(\beta_{\ell q} d_\ell) \cos\left[\frac{q\pi(x - x_\ell)}{w_\ell}\right]. \quad (\text{S7})$$

The exterior reflected contribution has $\rho\omega v_{y,n} = k_{y,n} A_n^{\text{dr}}$, while the incident contribution has $\rho\omega v_{y,\text{inc}} = -k_{y,\text{inc}} p_{\text{inc}}$.

Block operator

Let $\mathbf{B}$ project external pressure onto aperture cosine modes and let $\mathbf{D}$ project aperture cosine velocities onto global Floquet tests. If $\mathbf{C}_d = \text{diag}[\cos(\beta_{\ell q} d_\ell)]$ and $\mathbf{V}_d = \text{diag}[\beta_{\ell q} \sin(\beta_{\ell q} d_\ell)]$, the homogeneous pressure-velocity system can be written

$$\mathbf{F} \begin{bmatrix} \mathbf{A}^{\text{hom}} \\ \mathbf{C} \end{bmatrix} = 0, \quad \mathbf{F} = \begin{bmatrix} \mathbf{B} & -\mathbf{C}_d \\ \text{diag}(k_{y,n}/k_0) & i\mathbf{D}\mathbf{V}_d \end{bmatrix}. \quad (\text{S8})$$

The top rows enforce pressure continuity in each aperture. The lower rows enforce normal-velocity continuity after the rigid portions of the surface are included as zero-velocity regions. Keeping $C_{\ell q}$ as independent unknowns avoids the tangent poles that appear if the groove field is first eliminated in a driven calculation.

For the forced problem, the same continuity equations can be reduced to

$$\begin{aligned} [\text{diag}(k_{y,n}) - \mathbf{S}\mathbf{B}] \mathbf{A}^{\text{dr}} &= k_{y,\text{inc}} \mathbf{e}_0 + \mathbf{S}\mathbf{b}_{\text{inc}}, \\ \mathbf{S} &= \mathbf{D} \text{diag}[-i\beta_{\ell q} \tan(\beta_{\ell q} d_\ell)]. \end{aligned} \quad (\text{S9})$$

The vector in Eq. (S9) contains the incident aperture projections. A driven A_0^{dr} therefore includes the background reflection and is not the homogeneous A_0^{hom} used below.

STRICT ZERO AMPLITUDES AND SQUARE INTEGRABILITY

The normal-power efficiency of a propagating driven order is

$$\eta_n = \frac{\Re k_{y,n}}{k_{y,\text{inc}}} \left| \frac{A_n^{\text{dr}}}{A_{\text{inc}}} \right|^2, \quad A_{\text{inc}} = 1. \quad (\text{S10})$$

This expression alone cannot certify a BIC at a threshold. To see why, take one exterior term $p_n = A_n^{\text{hom}} e^{ik_{x,n}x - ik_{y,n}y}$ and integrate its pressure norm from the surface to height H . For a threshold order, $k_{y,n} = 0$ and

$$\int_0^H |p_n|^2 dy = H |A_n^{\text{hom}}|^2. \quad (\text{S11})$$

The same linear growth occurs for a propagating order after averaging over oscillations. For an evanescent order with $k_{y,n} = -i\alpha_n$ and $\alpha_n > 0$, the integral saturates as

$$\int_0^\infty |p_n|^2 dy = \frac{|A_n^{\text{hom}}|^2}{2\alpha_n}. \quad (\text{S12})$$

Consequently, at the positive-angle design point the strict condition is

$$A_{-1}^{\text{hom}} = 0, \quad A_0^{\text{hom}} = 0, \quad (\text{S13})$$

where $n = -1$ is the selected RA order and $n = 0$ is the finite-flux open order. At the time-reversed point the target is $n = +1$ and the second condition remains $A_0^{\text{hom}} = 0$. Zero grazing flux without $A_{n_{\text{RA}}}^{\text{hom}} = 0$ is therefore insufficient.

Radiation-reduced SVD

Let $\mathbf{F}_{\text{red}}$ be obtained from $\mathbf{F}$ by removing the columns corresponding to the threshold amplitude and all finite-flux open amplitudes. The remaining groove columns are retained. To avoid a spurious minimum caused by disparate modal scales, define

$$\begin{aligned} \mathbf{R} &= \text{diag}(r_i), \quad r_i = \max_j |F_{\text{red},ij}|_2, \\ \mathbf{C} &= \text{diag}(c_j), \quad c_j = \max_i |[R^{-1}\mathbf{F}_{\text{red}}]_{ij}|_2, \end{aligned} \quad (\text{S14})$$

and

$$\tilde{\mathbf{F}} = \mathbf{R}^{-1}\mathbf{F}_{\text{red}}\mathbf{C}^{-1}. \quad (\text{S15})$$

If $\mathbf{y}$ is the right singular vector associated with the smallest singular value of $\tilde{\mathbf{F}}$, the physical reduced vector is $\mathbf{u} = \mathbf{C}^{-1}\mathbf{y}$. The deleted amplitudes are reinserted as exact zeros to form $\mathbf{A}^{\text{hom}}$. We report both

$$\sigma_{\text{rel}} = \frac{\sigma_{\min}}{\sigma_{\max}}, \quad r_{\text{raw}} = \frac{\|\mathbf{F}_{\text{red}}\mathbf{u}\|_2}{\|\mathbf{u}\|_2}. \quad (\text{S16})$$

At $(N, K) = (313, 39)$ these are 1.2604×10^{-15} and 1.3766×10^{-13} , respectively. The two residuals probe different coordinates and should not be interpreted as two independent physical observables.

POLE CONTINUATION AND ITS CAVEATS

The complex pole is continued from the Rayleigh point by specifying the Riemann sheet of the selected order. For the positive-angle design the target is $n = -1$; the non-target open orders remain on the outgoing sheet and closed orders remain on the decaying sheet. A local complex coordinate $\bar{q}$ is used for the target normal wavenumber, and the pole frequency is reconstructed from

$$\Omega_p = \sqrt{(\kappa - 1)^2 + \bar{q}^2}, \quad Q = \frac{\Re \Omega_p}{2|\Im \Omega_p|}. \quad (\text{S17})$$

The continuation used for the reported local fit retains $(N, K) = (121, 15)$. It reaches a nearest finite Q of about 2×10^{12} before double-precision and sheet-tracking errors become material. The local fit is approximately quadratic, $Q \propto |\Delta\kappa|^{-2.00}$, over the resolved interval. This is a finite-model numerical trend, not a universal exponent and not a theorem about the continuum problem. The pole diagnostic is also not mixed with the $(313, 39)$ strict-root residual in Eq. (S16).

SINGLE-CAVITY ABLATION AND INTERPRETATION

The two-cavity design was compared with a single large rectangular cavity while retaining additional transverse cosine modes. In the searched rectangular-groove family, the best complete strict residual of the one-cavity model is approximately 0.124, whereas the two-cavity model reaches the residual in Eq. (S16). The result means that additional transverse modes in that one cavity did not supply the independent phase needed to cancel both A_0^{hom} and A_{-1}^{hom} . It is not a mathematical no-go theorem for a different one-cavity shape, a compound cavity, or a model with an additional internal resonator.

The mechanism can be written as two radiation phasor equations,

$$A_m^{\text{hom}} = \sum_{\ell, q} \mathcal{R}_{m, \ell q} C_{\ell q} = 0, \quad m \in \{0, -1\}, \quad (\text{S18})$$

where the wide cavity mainly supplies stored energy and the smaller cavity provides an independent phase trim. The roles are geometric and modal, not experimental assignments.

TABLE S1. Strict and driven convergence data retained for the theory draft. A dash means that a quantity was not used as the acceptance diagnostic at that truncation.

(N, K)	σ_{rel}	r_{raw}	f_{router} (kHz)	η_{anom}	role
(121, 15)	—	—	200.21475	0.30737	router retuning
(185, 23)	—	—	—	0.30662	router retuning
(281, 35)	—	—	—	0.30620	router retuning
(313, 39)	1.2604×10^{-15}	1.3766×10^{-13}	200.21687	0.30613	strict BIC / router

TABLE S2. Same-cell off-Rayleigh anomalous-reflection convergence. The target is $n = -1$ for positive incidence and $n = +1$ for negative incidence.

(N, K)	η_{target}	η_0
(81, 11)	0.97791	0.02209
(121, 15)	0.99298	0.00702
(161, 20)	0.99864	0.00136
(201, 25)	0.99963	0.00037
(313, 39)	0.99992	0.00008
(401, 50)	0.9996185	0.0003815

DIMENSIONAL AND ENVIRONMENTAL SENSITIVITY

The design scales with $a = \Omega c / f$. If the sound speed changes from 1500 m/s to 1480 m/s while the geometry and nominal frequency are held fixed, the same normalized feature is expected near $f \simeq 197.33$ kHz. This scaling is a guide for retuning, not a measured temperature calibration. A complete tolerance study should perturb d_1, d_2, w_1, w_2 , and g independently, recompute the strict residual and the driven efficiency, and report the matrix condition number alongside the efficiency. The near-BIC point at $\theta = 5.27044^\circ$ is especially ill-conditioned (cond $\simeq 1.6 \times 10^{10}$), so fixed-geometry efficiency is not a sufficient robustness metric there.

The off-Rayleigh operating point at $\theta = \pm 32.6328^\circ$ and $f = 202.430$ kHz is separated from this sensitivity discussion. Its active $n = -1$ threshold is approximately 141.825 kHz, so the 99.96185% target efficiency is not a Rayleigh-critical or BIC-enhanced value.

ENERGY, RECIPROCITY, AND REPRODUCIBILITY CHECKS

The Floquet normalization in Eq. (S10) makes the propagating-order basis power normalized. We use the ordered channel bases $(0, -1)$ at positive incidence and $(0, +1)$ at negative incidence. In the lossless model the

driven calculation obeys

$$\delta_E = \left| 1 - \sum_{n \in \text{prop}} \eta_n \right| \ll 1, \quad (\text{S19})$$

and the power-normalized two-channel matrix satisfies

$$\begin{aligned} S_{\pm}^\dagger S_{\pm} &= I, \\ S_+(\kappa) &= S_-^\dagger(-\kappa), \\ \eta_{-1}(+\theta) &= \eta_{+1}(-\theta). \end{aligned} \quad (\text{S20})$$

At the off-Rayleigh point, $\delta_E = 4.4 \times 10^{-16}$ at $(N, K) = (401, 50)$. This is a consistency check of the lossless driven solver, not evidence that losses vanish in a future finite sample.

At the positive-angle Rayleigh point, $k_{y,-1} = 0$. If $A_{-1}^{\text{hom}} \neq 0$, the threshold pressure is independent of y and its height-integrated norm grows linearly, as shown in Eq. (S11). The strict zero in Eq. (S13) removes this contribution; the retained $n = +1$ and higher orders are evanescent and have finite integrated norm. Complex conjugation and order reversal give the negative-angle state.

The nominal dimensional inputs are $f_0 = 200 \times 10^3$ Hz and $c = 1500$ m/s. The normalized design is converted using $a = \Omega c / f_0$ and $\sin \theta = \kappa / \Omega$. The external expansion retains an odd number N of orders centered on $n = 0$, and each groove retains K cosine modes. The strict result uses $(N, K) = (313, 39)$; the independent off-Rayleigh driven point uses $(401, 50)$. Before the singular-value decomposition, the strict operator is row- and column-scaled. The threshold and finite-flux columns are removed, the smallest right singular vector is reconstructed in physical coordinates, and both the scaled ratio and raw unscaled residual are recorded. Pole continuation uses the selected outgoing/decaying Riemann sheet. Driven efficiencies and energy closure are evaluated over all propagating orders retained by the external truncation.

PROPOSED FUTURE WATER-TANK EXPERIMENT

This section is a proposed protocol for a later paper. It records planned controls and observables only; no experiment has been performed for the present theory draft.

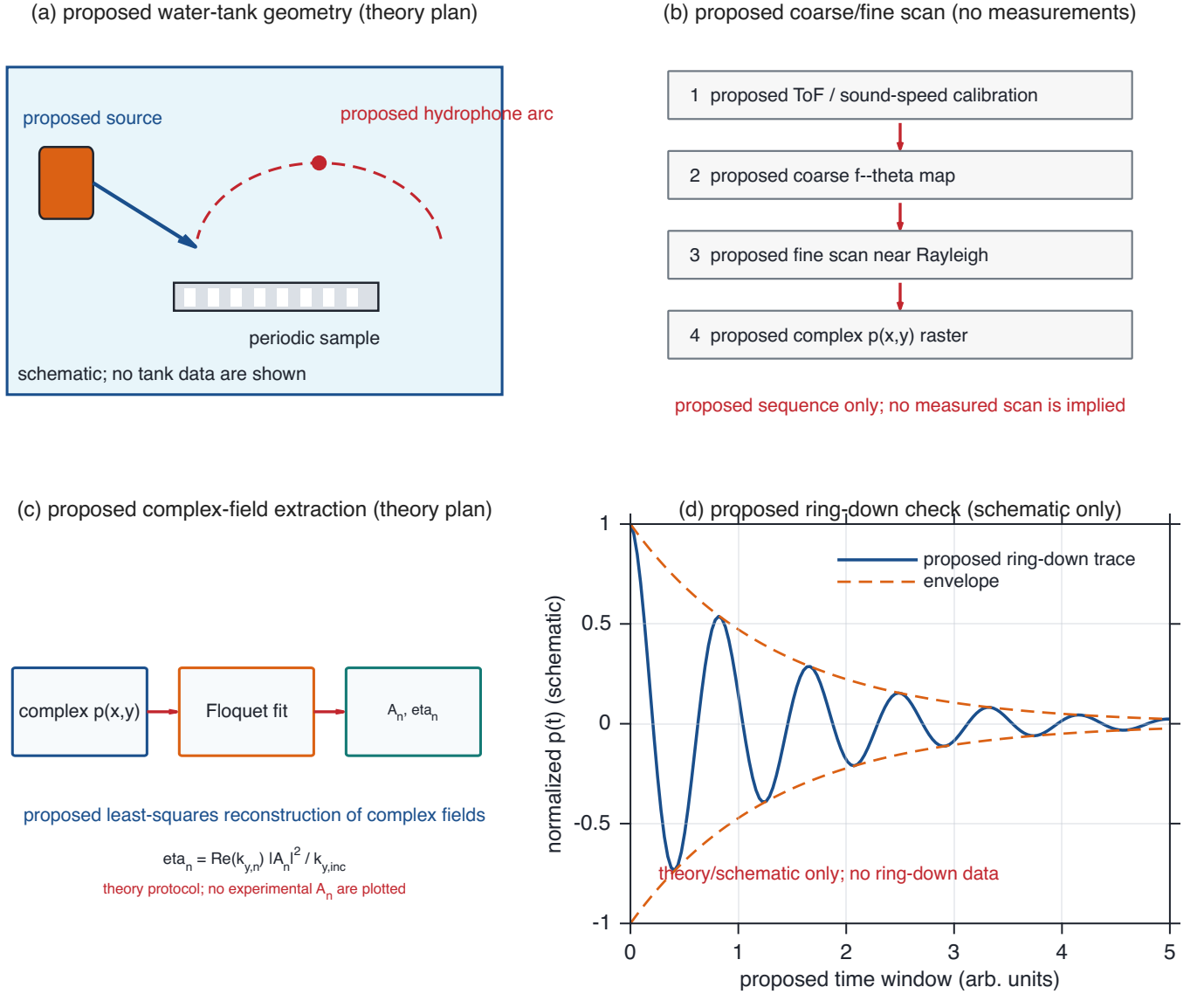


FIG. S1. PROPOSED future water-tank protocol, shown for planning only. (a) Tank geometry and the calibrated flat-plate reference. (b) The proposed angle–frequency scan sequence. (c) Complex near-field sampling followed by Floquet-amplitude extraction. (d) The proposed ring-down check. All panels are schematics or theory plans; no measurement is reported.

Samples and calibration

The proposed sample set contains an infinite-period surrogate in simulation, finite two-groove arrays with $M = 20, 40$, and 60 periods, a flat rigid reference plate, a single-wide-cavity control, and a full sample with the small cavity blocked. The material, backing thickness, and fluid–structure compliance must be measured or modeled before calling a sample rigid. The water temperature and sound speed should be recorded by time of flight. The source angular spectrum, hydrophone transfer function, and flat-plate complex reflection should be calibrated before scanning the grating.

Scan and field extraction

The proposed scan first maps approximately 190–205 kHz over $-7^\circ \leq \theta \leq 7^\circ$, then refines the features near $\pm 5.25^\circ$ and $\pm 6^\circ$. An independent map near $\pm 32.63^\circ$ should test the off-Rayleigh point. Complex pressure can be sampled over several periods at a set height y_0 and fitted to

$$p_{\text{sc}}(x, y_0) = \sum_n \tilde{A}_n(y_0) e^{ik_{x,n}x}, \quad (\text{S21})$$

$$A_n = \tilde{A}_n(y_0) e^{+ik_{y,n}y_0},$$

then inserted into Eq. (S10). The fit should use a measured incident field and a flat-plate reference so that the

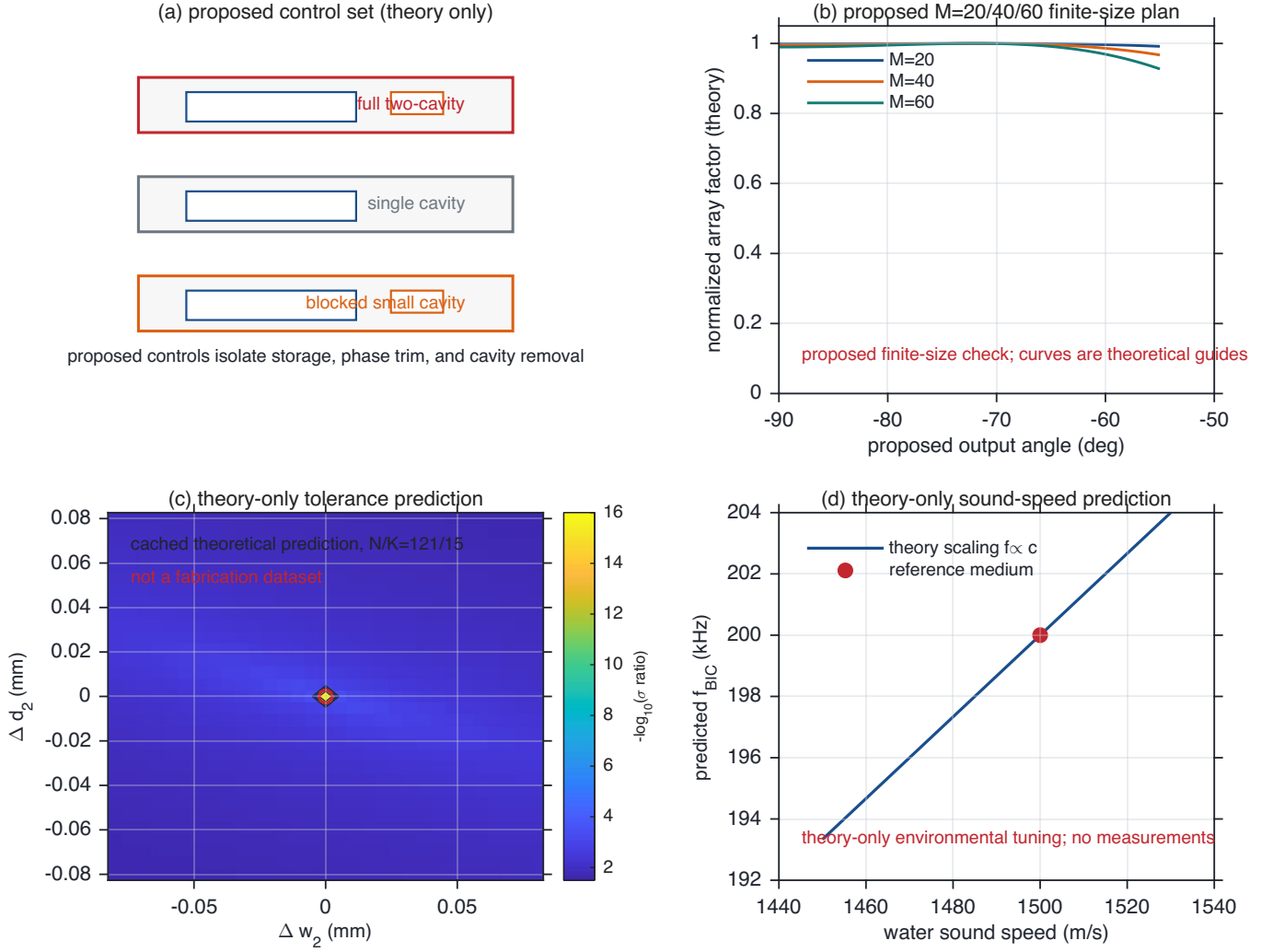


FIG. S2. PROPOSED controls and sensitivity plan. (a) Full, single-wide-cavity, and blocked-small-cavity controls. (b) The finite-array plan with $M = 20, 40$, and 60 periods. (c) The cached theory-only dimensional-tolerance prediction. (d) The theory-only sound-speed scaling prediction. No control measurement or completed tolerance map is claimed.

driven A_0^{dr} is not confused with A_0^{hom} . Near-field scans and far-field angular scans should be analyzed together, because a grazing pressure feature can carry little normal flux.

Ring-down and falsifiable checks

For a narrowband burst, switching off the source and fitting an amplitude decay $\exp(-t/\tau_A)$ would give the operational estimate

$$Q_{\text{ring}} = \pi f \tau_A. \quad (\text{S22})$$

The decisive test would be the joint convergence of the extracted complex Floquet amplitudes, the threshold-order pressure, the finite-flux power, and the reciprocal signed-order pair. A driven reflection dip alone would not validate the BIC. Finite width, source aperture, disorder,

viscous loss, and fluid–structure motion must be included in the uncertainty budget.

DATA AND CODE STATEMENT

The theory values are reproducible from the modal-matching implementation in the repository’s ‘Ni2019_MATLAB/’ directory. The relevant routines construct the driven modal solver, the pole-free full homogeneous operator, the strict Rayleigh reduced operator, the complex-pole continuation, the reciprocal direction pair, and the routing objective. The present supplement records the equations and numerical settings needed to reimplement those calculations; it does not depend on a finite-element package. All proposed experiment figures in Figs. S1 and S2 are marked as proposed and must not be read as completed measurements.